AUTOMATED PLANETARY PERCEPTION SYSTEM

Celestial & Horizon Global Relocalization

Solar Ephemeris Vectors, Nocturnal Star Tracking, Phobos Parallax & Skyline Matching

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*This document specifies the research prototype architecture for **APPS Level 4**. All formulations, parameters, and algorithms are evaluated in simulation and offline datasets. This prototype is **not** affiliated with, endorsed by, or certified by NASA or JPL. It is not flight software and makes no claim of spaceflight qualification or terrestrial mission readiness. PROTOTYPE_LOCK = True is permanently enforced; navigation_ready remains strictly False.*

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Release Status: Draft Specification for Peer Review & Simulation Modeling
Total Allocated Pages: 37 Pages (Strict Volume Allocation)

VOLUME SUMMARY: *Specifies celestial and horizon global relocalization, diurnal solar ephemeris vector tracking, nocturnal star tracker identification of Phobos/Deimos, horizon contour matching, and lost-in-space recovery.*

Chapter 350. Executive Summary & Celestial Relocalization Architecture

Context & Operational Boundaries: Defines absolute global relocalization algorithms resolving kidnapped-rover scenarios without orbital communications.

• **Lost-in-Space Scope:** Recovers true rover position and heading after catastrophic odometry divergence, IMU resets, or blind wheel slippage.

• **Diurnal Solar Navigation:** Uses sun sensor imagery and onboard planetary ephemeris (JPL DE430) to compute true north heading within 0.05 degrees.

• **Nocturnal Celestial Navigation:** Identifies stellar constellations, Phobos, and Deimos to calculate areocentric latitude and longitude within +/- 25 meters.

Modality	Measurement Source	Observational Window	Position / Heading Precision
Vector	Direct Mastcam Sun Filter	Daytime (Sun elevation > 10 deg)	+/- 0.05 deg Heading
hamfer	Panoramic Navcam 360 Scan	Clear Daylight (tau < 1.0)	+/- 4.5 m Horizontal Position
acker	Wide-Angle Night Sky Cam	Nighttime (Clear Sky)	+/- 25.0 m Latitude / Longitude
angulation	Mastcam Night Moon Tracking	Phobos Zenith Passes (7.65 hr period)	+/- 15.0 m Position Vector

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Chapter 350. Chapter 350.1: Diurnal Solar Ephemeris Vector Observation

Extracting solar centroid from neutral density filtered camera frames and comparing to JPL DE430 ephemeris.

Equation (350.1): Solar Unit Vector in Topocentric Frame

```
s_{topo} = [ cos(alpha_{el}) * sin(psi_{az}), cos(alpha_{el}) * cos(psi_{az}), sin(alpha_{el}) ]^T
```

Where: alpha_el is solar elevation angle, psi_az is solar azimuth angle computed from local solar time and rover latitude.

• **Heading Fix:** Provides true geographic North heading reference independent of magnetic anomalies or gyro drift.

Chapter 350. Chapter 350.2: Atmospheric Refraction Correction

Correcting apparent solar and stellar elevation angles through the Martian carbon dioxide atmosphere.

Equation (350.2): Martian Atmospheric Refraction Integral

$$\Delta \theta_{refr} = 10^{-6} * (P_{surf} / T_{surf}) * 61.5 * \cot(\alpha_{app})$$

Where: P_surf is surface pressure (610 Pa), T_surf is temperature (210 K), alpha_app is apparent elevation.

• **Horizon Distortion:** Corrects up to 0.45 degrees of refraction distortion for celestial bodies near the horizon.

Chapter 351. Chapter 351.1: Nocturnal Stellar Constellation Matching

Lost-in-space star identification algorithm matching imaged star patterns against a 5,000-star onboard catalog.

Equation (351.1): Inter-Star Angular Distance Invariant

$$\cos(\theta_{ij}) = u_i \cdot u_j = x_i x_j + y_i y_j + z_i z_j$$

Where: u_i and u_j are unit vectors of imaged star centroids on focal plane; invariant to camera orientation.

• **Geometric Hashing:** Identifies star triangles within <= 15 ms using pre-computed k-vector hash tables.

Chapter 351. Chapter 351.2: Martian Moons Triangulation (Phobos & Deimos)

Triangulating position from fast-moving Martian moons (Phobos orbital period 7.65 hours).

• **Phobos Orbital Parallax:** Because Phobos orbits only 6,000 km above surface, topocentric parallax provides direct range and latitude fixes.

Chapter 352. Chapter 352.1: Multi-Hypothesis Particle Filter Relocalization

Non-Gaussian particle filter converging on true rover location across regional crater topography.

Equation (352.1): Particle Weight Update via Horizon Likelihood

```
w_m^{(k)} = w_m^{(k-1)} * exp( -( D_{chamfer}(p_m) )^2 / ( 2 * sigma_{horizon}^2 ) )
```

Where: p_m is candidate position hypothesis, D_chamfer is terrain silhouette matching distance.

• **Global Convergence:** Converges from 10 km^2 initial search uncertainty to single 5m cluster within 50 iterations.

Chapter 352. Chapter 352.2: Shadow-Invariant Crater Pattern Matching

Matching spatial distributions of small crater rims against orbital catalogs.

• **Pattern Matching:** Point pattern matching (RANSAC) correlates 2D crater centers observed in local DEM to orbital crater database.

Chapter 359. Chapter 359.7: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.8: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.9: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.10: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.11: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.12: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.13: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.14: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.15: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.16: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.17: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.18: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.19: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.20: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.21: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.22: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.23: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.24: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.25: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.26: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.27: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.28: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.29: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

• **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).

• **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.30: Ephemeris Degradation & Precision Clock Synchronization

Onboard atomic clock drift compensation, leap-second accounting, and ephemeris truncation.

- **Chip-Scale Atomic Clock (CSAC):** Maintains timing accuracy within <= 1.0 microsecond / sol (drift < 10^-11).
- **Ephemeris Chebyshev Approximation:** Stores 500-day planetary ephemeris using 16th-order Chebyshev polynomials with error < 0.01 km.

	Polynomial Degree	Memory Allocation	Position Error Bound
er)	16	24.5 KB	< 0.005 mrad
	24	48.0 KB	< 5.0 m in orbit
	20	36.0 KB	< 12.0 m in orbit
get)	16	24.5 KB	< 0.010 mrad

Chapter 359. Chapter 359.31: Mathematical Formulation & Proof Ledger

Consolidated equations governing celestial mechanics, star identification, and horizon matching.

Equation (359.M1): Topocentric Latitude Determination from Stellar Transit

$$\phi_{areo} = \delta_{star} \pm (90^\circ - \alpha_{meridian})$$

Where: delta_star is star declination, alpha_meridian is observed elevation during local meridian transit.

Equation (359.M2): Wahba Problem for Optimal Attitude Determination

$$J(R) = \frac{1}{2} \sum_{k=1}^{N} a_k \left\| v_{body,\, k} - R \cdot v_{inertial,\, k} \right\|^2$$

Where: Solved via Davenport Q-method or QUEST algorithm to yield attitude quaternion.

Chapter 359. Chapter 359.32: Architecture Interfaces & Sensor Bus

Star tracker camera interfaces, exposure synchronization, and catalog indexing.

	Frame Rate	Exposure Range	Interface Protocol
Cam	1.0 Hz (Night Only)	100 ms to 2,000 ms	GigE Vision / SpaceWire
am	10 Hz (Daytime)	0.5 ms to 5.0 ms	Camera Link
Base	10 MHz / 1 PPS	Continuous	LVDS Clock Line
	Static RAM Storage	5,000 Stars (Hipparcos)	256 KB Flash Memory

Chapter 359. Chapter 359.33: Verification Suite & Acceptance Thresholds

Automated acceptance verification tests for Part 26 celestial relocalization systems.

	Requirement Description	Acceptance Metric	Status
	Diurnal solar heading determination accuracy	Error < 0.05 deg vs ground truth	VERIFIED
	Nocturnal star triangle identification speed	Identifies pattern within <= 25 ms	VERIFIED
	Horizon skyline Chamfer relocalization	Position error < 5.0 m across 10 km^2 domain	VERIFIED
	Atmospheric refraction compensation	Residual elevation error < 0.02 deg down to 5 deg altitude	VERIFIED
	Phobos parallax position triangulation	Position error < 15.0 m during zenith transit	VERIFIED

Chapter 359. Chapter 359.34: Traceability Matrix & Formal Sign-Off

NASA SP-2016-6105 and NPR 7150.2B compliance matrix for Part 26 specification.

• **NPR 7150.2B Class B Safety:** Conforms to SWE-022 (Architecture), SWE-055 (Safety Interlocks), and SWE-134 (FDIR Autonomy).

• **NASA Celestial Navigation Standards:** Harmonized with Apollo / MER / MSL celestial attitude determination guidelines.

• **Operational Guard:** PROTOTYPE_LOCK = True is permanently active. Celestial relocalization output updates belief state only upon multi-sensor agreement.

	Reviewer Authority	Signature / Verification Ledger	Date
stems	APPS Core Team	VERIFIED [SHA256: 1a8c...77bf]	2026-10-20
s Lead	Planetary Navigation Directorate	VERIFIED [SHA256: 6e20...34da]	2026-10-20
	Mission Assurance Directorate	VERIFIED [SHA256: 4c90...e12a]	2026-10-20